

Data Science Lab Experiment 5

Experiment Title: Programs to Handle Data Using Pandas

Question

Write a Python program using the **Pandas** library to perform the following tasks:

1. Create a **Pandas Series** containing the values **1, 3, 5, 6, and 8**, and display it.
2. Create a **DataFrame** from a dictionary containing the details of **Country, Capital, Area, and Population** for the BRICS countries, and set the row labels as **BR, RU, IN, CH, and SA**.
3. Read the dataset **cars1.csv** into a DataFrame and:
 - Display the complete dataset.
 - Display the first four records.
 - Display the fifth and sixth records.
 - Display the third row using the **iloc[]** method.
4. Create a DataFrame containing the columns **Name, Gender, and Age**, and perform DataFrame slicing to:
 - Display the first two rows.
 - Display all rows from the third row onwards.
5. Create a Pandas Series containing **four random numbers**, and display:
 - The complete Series.
 - The values stored in the Series.
 - The first two elements using **head()**.
 - The last three elements using **tail()**.
6. Create a DataFrame containing the columns **Name, Age, and Rating**, and display the following DataFrame properties:
 - Transpose of the DataFrame.
 - Row and column labels using **axes**.
 - Data type of each column using **dtypes**.
 - Whether the DataFrame is empty using **empty**.
 - Number of dimensions using **ndim**.
 - Shape of the DataFrame using **shape**.
 - Total number of elements using **size**.
 - Values stored in the DataFrame using **values**.
 - Number of missing values in each column using **isnull().sum()**.

